25900156 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCPHY100 - OBSERVATIONAL ASTRONOMY

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the approximate age of the Universe in years according to the Big Bang theory? [K] / [CO1]

   a). 4.6 billion b). 10 billion c). 13.8 billion d). 20 billion

2. Which of the following is a unit used to express the size of our solar system? [K] / [CO1]

   a). Parsecs b). Astronomical Unit (AU)

   c). Light years d). Kilometers

3. Which planet is closest to sun in heliocentric model? [K] / [CO1]

   a). Earth b). Mars c). Mercury d). Venus

4. What is the duration of one Solar day? [K] / [CO4]

a). 12 hours b). 24 hours c). 48 hours d). 365 days

5. The apparent path of the Sun against the background stars is known as: [K] / [CO4]

   a). Equator b). Ecliptic c). Elliptic d). Zodiac

6. Which part of a refractive telescope magnifies the image? [K] / [CO2]

   a). Primary Mirror b). Objective Lens

   c). Eyepiece Lens d). Diagonal Mirror

7. Where is the location of James web Telescope? [K] / [CO2]

   a). 2000 km from earth b). At Mars

   c). Approximately 15 lakh km from earth d). Close to Sun

8. Magnifying Power of a telescope is defined as the ratio of: [K] / [CO2]

   a). Focal length of Objective Lens to Focal length of Eyepiece Lens

   b). Focal length of Eyepiece Lens to Focal length of Objective Lens

   c). Radius of Eyepiece Lens to Radius of Objective Lens

   d). Radius of Objective Lens to Radius of Eyepiece Lens

9. Eventually what does a white dwarf become? [K] / [CO3]

   a). A pulsar b). A neutron star

   c). A black hole d). A black dwarf

10. A collapsed core of a massive star after supernova is called as: [K] / [CO3]

    a). Pulsar b). White Dwarf

    c). Comet d). Neutron Star

11. What is the most famous nebula visible to the naked eye, known for star formation? [K] / [CO3]

    a). Orion Nebula b). Crab Nebule

    c). Helix Nebula d). Ring Nebula

12. Who classified galaxies based on their shapes? [K] / [CO4]

a). James Webb b). Edwin Hubble

c). Carl Sagan d). Copernicus

13. When does a full lunar eclipse occur? [K] / [CO5]

a). Once in a month

b). When earth passes directly in front of new moon

c). When Sun passes directly in front of new moon

d). When earth passes directly in front of full moon

14. Which is the nearest planet to sun? [K] / [CO5]

a). Mercury b). Mars

c). Venus d). Jupiter

15. Where is the location of main Asteroid belt in the Solar System? [K] / [CO5]

a). Beyond Neptune b). Between Mars and Jupiter

c). Near Earth's orbit d). Within Saturn's rings

[1x15 = 15]

Part B

Multiple Choice Questions

Answer all questions

Each question carries **2** marks

16. Which method was used by Galileo Galilei to test hypotheses? [K] / [CO1]

a). Philosophical reasoning

b). Thought experiments only

c). Systematic observation and experimentation

d). Religious doctrine

17. What does the term "deferent" refer to in the Ptolemaic model? [K] / [CO1]

a). The small circular orbit of a planet

b). The retrograde motion of planets

c). The larger circle along which the center of the epicycle moves

d). The center of the universe

18. What is the brightest star in the constellation of Orion? [K] / [CO4]

a). Betelgeuse b). Rigel

c). Sirius d). None of these

19. Which electromagnetic radiation is used to detect quasars and pulsars? [K] / [CO4]

a). Gamma Rays b). Radio Waves

c). Visible Light d). Infra Red Rays

20. How often does Hubble Space Telescope orbits earth? [K] / [CO2]

a). Every 30 mins b). Every 60 mins

c). Every 90 mins d). Every 120 mins

21. Which colour is most associated with the hottest stars? [K] / [CO3]

a). Red b). Blue c). Yellow d). White

22. Which of the following best describes the core condition of a red giant star? [U] / [CO3]

a). The core is composed of hydrogen and continues hydrogen fusion

b). The core is composed of helium and contracts while the outer layers expand

c). The core expands and cools, reducing nuclear fusion

d). The core remains unchanged, but the outer layers shrink

23. At what approximate speed does the Sun revolves around the centre of Milky way galaxy? [K] / [CO4]

a). 250 m/s b). 250 km/s

c). 250 km/hr d). 25 km/hr

24. Why doesn't solar eclipse occur every month? [K] / [CO5]

a). The Sun's orbit is tilted b). The Earth's orbit is tilted

c). The Moon's orbit is tilted d). The planets' orbits are tilted

25. Which of the statement about sun is wrong? [U] / [CO5]

a). It is the star at the centre of solar system

b). It is spherical in shape

c). About 150 million Km far from earth

d). Main component of sun is helium

[2x10 = 20]

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Answer all questions

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[1x15 = 15]

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Multiple Choice Questions

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